

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First/Second Semester of B. Tech. (CE/IT/EC/CL/ME/EE) Theory Examination****November - December 2018****ME101.01/ME101 Engineering Graphics****Date: 26/11/2018, Monday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and Section II must be attempted in separate drawing sheets.
3. Make suitable assumptions and draw neat figures wherever is required.
4. Figures drawn in the question paper are not to the scale.
5. Show all the necessary dimensions in the solution.
6. Figures to right indicate the full marks.
7. Stencil work is not compulsory.

SECTION – I**Q - 1 Do as directed.**

1. What is the difference between sketching and engineering drawing? [01]
2. Protractor is used to draw or measure: [01]
 - a) Angles
 - b) Curves
 - c) Circle
 - d) None of these
3. 1 meter = _____ centimeters. [01]
 - a) 10
 - b) 100
 - c) 1000
 - d) 10000
4. The front view of a point is observed above XY line, when the point is in _____. [01]
 - a) First quadrant
 - b) Second quadrant
 - c) Both a & b
 - d) None of these
5. Differentiate between first angle projection system and third angle projection system. [02]
6. The front view of a line is observed as a point, when a line is perpendicular to VP. True or False? [01]

Q – 2.a Draw a parabola using any method. Take base and axis of parabola as 140 mm and 100 mm respectively. Draw normal and tangent at any point on the curve. [06]

Q – 2.b A line BC has one end B in VP and other end C is 25 mm above HP. Draw the projection of the line BC when it is inclined 40° to HP and 50° to VP. Take the length of BC as 60 mm. [06]

OR

Q – 2.b A line XY is 75 mm long and it is 30° & 40° inclined to HP & VP respectively. The end X is 12 mm above HP and 10 mm in front of VP. Draw projection of line XY. A line is in 1st quadrant. [06]

Q – 2.c Draw the projection of point S when it is in 1st quadrant, 2nd quadrant, 3rd quadrant and 4th [04]

quadrant. A point S is 10 mm away from HP and 15 mm away from VP.

Q – 3.a **Figure 1** shows pictorial view of an object. Draw the following views: (1) Front View, and [08]
(2) Left Hand Side View. Dimensions shown in the figure are in mm. (**Figure 1** is on Page 3)

Q – 3.b What is AutoCAD? Discuss it in detail. [04]

SECTION – II

Q - 4 Do as directed.

1. If the length and width of a cube is 50 mm, then what is the height of cube? [01]
a) 40 mm b) 50 mm c) 60 mm d) 70 mm
2. Microsoft Word is a CAD software. True or False? [01]
3. The front view of square plate is a _____, when it is parallel to VP. [01]
a) Square b) Line c) Point d) Circle
4. The four-center method is used to draw isometric drawing of circle. True or False? [01]
5. Development of a square prism gives _____ rectangles. [01]
a) 1 b) 2 c) 3 d) 4
6. Enlist the methods of development of solid. [01]
7. Development of a cylinder is an ellipse. True or False? [01]

Q – 5.a A pentagonal plate of side 30 mm is resting on one of its corner on HP. Opposite side to this corner is 20 mm above HP and inclined at 40° to VP. Draw the projections and find inclination of the plate with HP. [06]

Q – 5.b A cylinder has base diameter of 60 mm and axis 75 mm, is resting on one of its generators on the H.P. The top view of the axis of cylinder makes an angle of 30° with the V.P. Draw the projections of the cylinder. [08]

OR

Q – 5.b Classify the solids with necessary sketches. [08]

Q – 6.a A square pyramid of side of base 40 mm and height 75 mm is resting with its base on H.P. [08]
Draw the isometric drawing of the square pyramid.

OR

Q – 6.a **Figure 2** shows the orthographic projections of an object in first angle projection system. [08]

Draw its isometric view. Dimensions shown in the figure are in mm. (**Figure 2** is on Page 3)

- Q – 6.b** A square prism has edge of base as 40 mm and height of axis as 70 mm, is resting on HP [06]
on its base. Draw the development of a square prism.

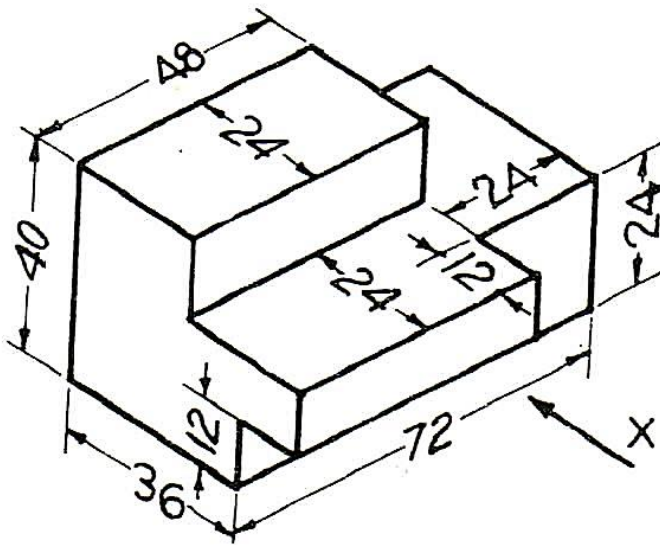
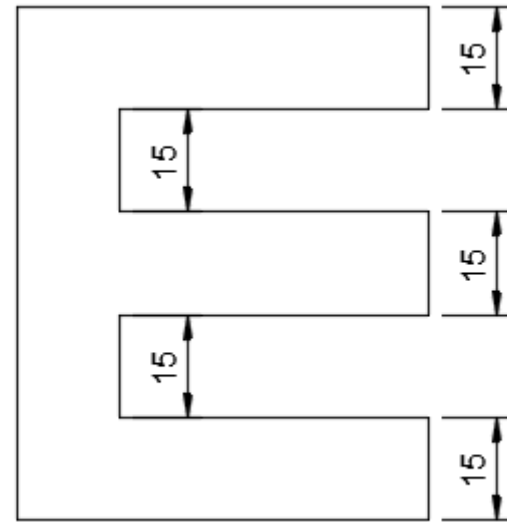
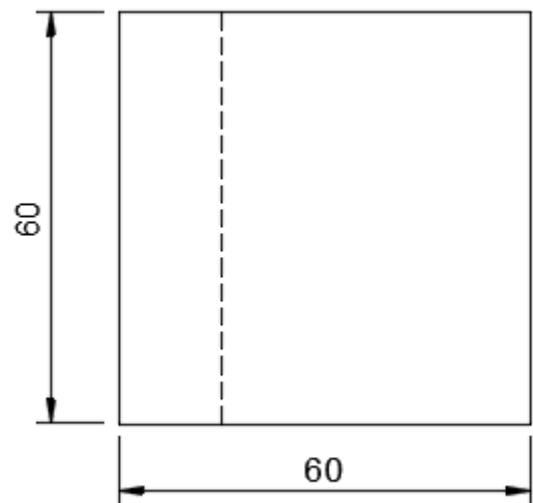


Figure 1



FV



TV

Figure 2
